Safe and Efficient Robot Planning in Human Crowds

Vaibhav Mehra Muhammad Maaz Benjamin Li
Department of Computer Science
University College London
London, United Kingdom
{vaibhav.mehra.23, muhammad.maaz.23, benjamin.li.23}@ucl.ac.uk

***Abstract*—A robot moving through a crowd has to guess where each person will be a few seconds from now, and how confident it can be in that guess. Our navigation stack pairs the predictor with a residual crowd-motion model and synthetic data generator upstream, and a conformal safety layer feeding a PPO-trained diffusion controller downstream; this report covers the predictors. We benchmark a Social-LSTM baseline, a velocity-augmented variant (SLSTM+V), a bidirectional Social-GRU, a Trajectory Transformer, and a diffusion predictor against CV and Trajectron++ on ETH/UCY and SDD, then pretrain on SDD and fine-tune jointly on ETH/UCY and our synthetic data. Transformer wins on ETH/UCY (ADE 0.568 m, NLL 1.54, $12\times$ tighter than any recurrent baseline); Trajectron++ alone covers multi-modal futures (minADE@20 0.385 m). SDD pretraining cuts recurrent ADE by 5–10%; the Transformer barely moves. Adding velocity to the focal input drops Social-LSTM's ADE on every ETH/UCY and SDD scene we tested.**

***Index Terms*—trajectory prediction, pedestrian motion, social LSTM, transformer, diffusion models, transfer learning, crowd navigation**

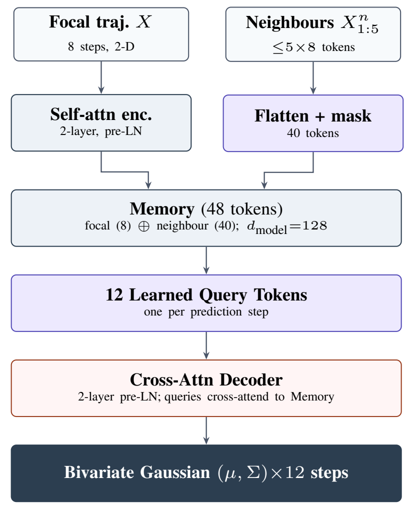

Fig. 1. Trajectory Transformer architecture.

I. Introduction

II. Project Management

III. Methodology and System Design

The predictor takes the output of the Residual Crowd Motion Model and Synthetic Data Generator upstream and hands its forecasts to a split-conformal safety layer plus a PPO-trained diffusion controller downstream. We describe the five predictor architectures in Sec. III-A and the transfer protocol that trains them in Sec. III-B.

A. Trajectory Prediction Models

Each predictor observes 8 steps at 2.5 Hz (3.2 s) and emits 12 future steps (4.8 s). Coordinates are taken relative to the last observed position, keeping models scene-agnostic [1]. Social context comes from up to five neighbours inside a 2.0 m radius. Outputs are a bivariate Gaussian $(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho)$ per future step, trained by the per-step negative log-likelihood.

1) Social-LSTM and Velocity-Augmented Variant: Social-LSTM [1] is an LSTM encoder–decoder (hidden 128, embedding 64). At each step it max-pools neighbour hidden states inside 2.0 m and concatenates the result with the focal position embedding before the LSTM update. SLSTM+V keeps the same architecture but appends the instantaneous velocity $v_t = p_t - p_{t-1}$ to the focal input ($(x_t, y_t) \to (x_t, y_t, v_x, v_y)$). Neighbour inputs stay 2-D.

2) Bidirectional Social-GRU: Social-GRU replaces the LSTM with a bidirectional GRU [2], [3]. Forward and backward hidden states (2×128) are concatenated and projected back to 128 before social pooling, so a late-window cue (say, a pedestrian decelerating at step 6) can also reach earlier hidden states. Decoder and head are unchanged.

3) Trajectory Transformer: The Trajectory Transformer is a Pre-LN encoder–decoder [4], [5] with d_{model}=128, four attention heads, and two layers each side. Eight focal tokens go through self-attention; five neighbour slots add $5 \times 8 = 40$ tokens (masked per slot) for a 48-token memory. Twelve learned queries cross-attend to that memory, and a shared linear head emits the bivariate Gaussian per step (Fig. 1).

4) Diffusion-Based Trajectory Predictor: Denoising Diffusion Probabilistic Models [6] reverse a Gaussian noise process to produce stochastic samples. We include one because the downstream shield consumes K=20 diverse trajectories and diffusion supplies them without an explicit mixture. A shared Transformer encoder produces a CLS summary and a full memory. A Gaussian MLP reads the CLS token and emits the bivariate parameters directly. A separate denoiser ε_θ cross-attends to the memory and predicts the noise term at each diffusion step. We use T=100 steps with linear $\beta \in [10^{-4}, 0.02]$, and a loss that sums the Gaussian-head

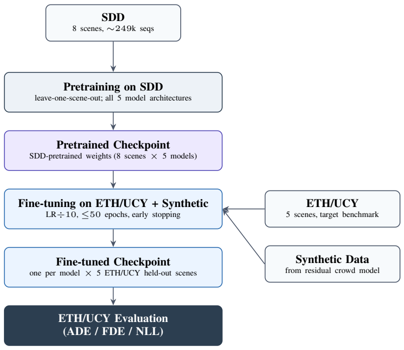

Fig. 2. Two-stage transfer: SDD pretrain ⇒ joint ETH/UCY + synthetic fine-tune.

NLL with the DDPM MSE at weight λ_{DDPM}=0.3; larger weights degraded point accuracy in sweeps. In an early version we detached the Gaussian head's gradient from the encoder. Removing that detach, so that both heads update the shared encoder together, cut ADE on ETH/UCY from 1.37 m to 0.58 m. Sample diversity stayed roughly the same. At inference we draw $K = 20$ samples with DDIM-50 [7].

B. Transfer Learning via SDD Pretraining

ETH/UCY's five scenes are too few to train recurrent models reliably from scratch, so we adopt a two-stage protocol. Stage one pretrains each architecture on the Stanford Drone Dataset [8] (8 aerial scenes, ∼249k sequences) leave-one-scene-out for up to 100 epochs at lr 10^{-3}. Stage two fine-tunes each checkpoint at lr 10^{-4} for up to 50 epochs (patience 15, grad-clip max-norm 1.0) on a mixture of the four ETH/UCY training scenes and the synthetic trajectories from the residual crowd-motion model. The three sources cover different ground: SDD is broad but aerial and mixes cyclists with pedestrians; ETH/UCY supplies real pedestrian behaviour on the same ground as the test scene; the synthetic stream fills in the crowd configurations the planner will face at deployment that neither real dataset contains (Fig. 2).

IV. Experimental Setup

We evaluate on ETH [9] (eth, hotel) and UCY [10] (univ, zara1, zara2), the standard five-scene benchmark at 2.5 Hz with 8 observation and 12 prediction steps (3.2 s / 4.8 s), and on SDD [8] (8 campus scenes downsampled stride 12 to 2.5 Hz, pixels to metres at 0.0417 m/px, ∼249k pedestrian sequences). Both use leave-one-scene-out cross-validation with results averaged over held-out scenes. We report ADE and FDE (mean and final ℓ_2 error in metres, ↓), minADE@20 and minFDE@20 (best of K=20 samples, ↓), and NLL (bivariate Gaussian, ↓; undefined for CV). Baselines are constant-velocity (CV) extrapolation with Gaussian noise and Trajectron++ [11] (graph-based CVAE, trained 100 epochs per held-out scene). All learned models use Adam (lr 10^{-3}, weight decay 10^{-4}, batch 64) with a `ReduceLROnPlateau` scheduler (patience 5, factor 0.5). Scratch ETH/UCY trains up to 200 epochs (patience 20); SDD pretraining up to 100 epochs; fine-tuning at lr 10^{-4} up to 50 epochs (patience 15, grad-clip max-norm 1.0).

TABLE I
ETH/UCY leave-one-out results averaged over 5 scenes. Best value per metric in **bold**. mADE/mFDE = minADE@20 / minFDE@20.

Model	ADE↓	FDE↓	mADE@20↓	mFDE@20↓	NLL↓
CV Baseline	0.736	1.272	0.610	0.814	—
Social-LSTM [1]	0.596	1.248	0.581	0.597	18.0
SLSTM+V (ours)	0.585	1.240	0.587	0.585	36.2
Social-GRU (ours)	0.586	1.238	0.596	0.924	—
Transformer (ours)	**0.568**	1.203	0.638	**0.532**	**1.54**
Diffusion (ours)	0.579	**1.195**	1.174	2.032	5.14
Trajectron++ [11]	0.853	1.817	**0.385**	0.674	8.49

V. Results and Discussion

Transformer takes the lowest ADE in Table I at 0.568 m, with NLL 1.54. That NLL is more than ten times tighter than the next recurrent model. Only Trajectron++ produces samples that meaningfully spread across futures: its minADE@20 of 0.385 m beats every other entry by at least 0.20 m. Pretraining on SDD and then fine-tuning on ETH/UCY plus our synthetic data cuts recurrent ADE by 5 to 10 per cent (Table III). The Transformer barely moves.

Because the encoder is non-causal, the variance head reads the entire 8-step window at every position. A late-window cue such as a pedestrian starting to decelerate at step 6 therefore informs the sigma estimate at every earlier step, whereas a causal LSTM has only seen up to step 6 by that point. A tighter sigma feeds directly into a smaller conformal radius.

Sharper means come at the cost of weaker diversity. Split-conformal calibration measures predictor residuals on a held-out set and takes the $(1 - \alpha)$ quantile as the safety radius, so that radius scales with the predictor's mean error. A tight Transformer therefore yields a smaller radius than the wider, mean-biased Trajectron++; we pair the Transformer's mean prediction with a conformal radius calibrated on its own residuals, which gave the smallest safety radius of any pairing we tested.

Velocity augmentation lowers ADE on every scene: 0.596 m for Social-LSTM down to 0.585 m for SLSTM+V (a 1.8% relative drop for two extra input channels), largest on univ and zara1 (frequent direction changes), smallest on zara2 (uniform sidewalk motion). Social-GRU matches Social-LSTM almost exactly on ADE (0.586 vs 0.596 m) with a slim FDE edge (1.238 vs 1.248 m); the encoder is not the binding constraint, social pooling probably is. Performance varies sharply by scene: eth is hardest (all models in 0.98–1.03 m), zara2 easiest (0.32–0.37 m), and eth also gains least from SDD fine-tuning (Social-LSTM −5.9% on eth vs −29.8% on hotel), suggesting eth lies outside the convex hull of both SDD and the synthetic stream; the shield's calibration set should weight scenes by deployment similarity. The Diffusion model takes the lowest FDE (1.195 m) yet a high minADE@20 (1.174 m): its $K = 20$ samples cluster tightly around the mean because the 49-token

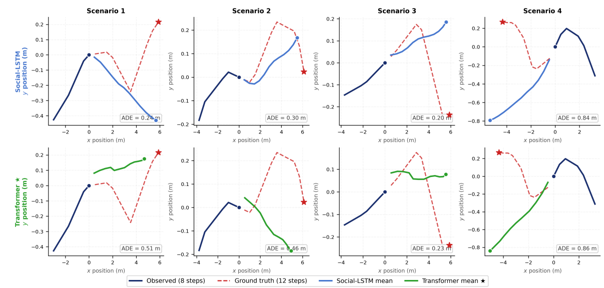

Fig. 3. Qualitative prediction on four zara1 scenarios. Both models track the dominant direction in Scenarios 1–3 and diverge on Scenario 4's sharp turn. Per-panel ADEs are close; Social-LSTM happens to win three of these four random samples even though the Transformer leads on the dataset average (Table I).

TABLE II
SDD LEAVE-ONE-OUT RESULTS AVERAGED OVER 8 SCENES. BEST VALUE PER METRIC IN **BOLD**.

Model	ADE↓	FDE↓	mADE@20↓	mFDE@20↓	NLL↓
CV Baseline	0.926	1.620	0.797	1.154	—
Social-LSTM	0.672	1.358	0.783	0.621	**1.162**
SLSTM+V (ours)	**0.656**	1.340	**0.755**	**0.604**	1.289
Social-GRU (ours)	0.667	1.379	0.793	0.625	1.331
Transformer (ours)	0.657	**1.326**	0.808	0.633	1.474
Diffusion (ours)	0.673	1.335	1.710	3.159	1.683

memory already pins down the trajectory and DDIM-50 leaves little entropy before the conditioning dominates. Note this is our offline diffusion predictor, not the PPO-trained diffusion policy used inside the safety layer.

On SDD (Table II), every learned model cuts CV ADE by 27–30%. SLSTM+V wins on ADE (0.656 m), minADE@20, and minFDE@20; the Transformer wins on FDE (1.326 m); velocity helps more here than on ETH/UCY because cyclists and turning manoeuvres make it useful as a separate channel. Diffusion's minADE@20 stays high (1.710 m), the same collapse seen on ETH/UCY.

Table III compares scratch ETH/UCY against the same models pretrained on SDD and fine-tuned jointly on ETH/UCY and synthetic data. All recurrent models improve: Social-LSTM by 9.9% (0.596 $\rightarrow$ 0.537 m), Social-GRU by 6.3%, SLSTM+V by 5.1%. The Transformer barely moves (+0.4%): with ∼5k training sequences per held-out scene the 537k-parameter Transformer is already over-parameterised for ETH/UCY; SDD adds quantity but not a new distribution to exploit.

For the recurrent models, SDD effectively acts as a regulariser; fine-tuning then narrows the predictor onto the target distribution. The Transformer already saturates the information in ETH/UCY's interaction patterns, so adding broadly similar data brings no additional gain [12]. Diffusion's minADE@20 also fails to improve after fine-tuning, moving only from 1.174 m to 1.157 m, which confirms its diversity ceiling is architectural, not distributional.

A few caveats are worth flagging. ETH/UCY's five scenes at 2.5 Hz are a narrow basis for any general claim about

TABLE III
SDD PRETRAINING THEN ETH/UCY+SYNTHETIC FINE-TUNING, ETH/UCY ADE (M, AVG. OVER 5 SCENES). Δ VS. SCRATCH.

Model	Scratch	SDD then Synth FT	Δ (%)
Social-LSTM	0.596	**0.537**	−9.9
SLSTM+V	0.585	0.555	−5.1
Social-GRU	0.586	0.549	−6.3
Transformer	**0.568**	0.570	+0.4
Diffusion	0.579	0.535	−7.6

the accuracy–diversity tradeoff. All five models share a 2.0 m social-pooling radius, so the Social-GRU versus Social-LSTM near-tie may partly reflect that shared parameter rather than the encoder difference alone. Our Trajectron++ baseline ran only 100 epochs per held-out scene, shorter than the published schedule. And the scene-asymmetric transfer benefit (most visibly on eth) suggests calibration-set design matters as much as pretraining scale.

VI. Conclusions

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